

A Blockchain-Enabled Traceability Platform for Authentic "Farm-to-Consumer" Supply Chain

Develop a low-cost, scalable blockchain solution to ensure transparent, tamper-proof tracking of agricultural produce from the farm to the consumer

❖ PROBLEM IDENTIFICATION

- **Lack of Transparency:** Consumers can't see food origins or handling, causing mistrust.
- **Farmer Exploitation:** Opaque supply chains lead to underpricing and exploitation.
- **Consumer Mistrust:** Adulteration and fake certifications reduce trust.
- **Data Tampering:** Records can be altered, enabling fake invoices and quality reports.

❖ PROPOSED SOLUTION

Farmer Interface: Registers harvest details (date, location, crop, certifications), generates a unique batch ID on the blockchain, and tracks transparent pricing to prevent exploitation.

Distributor Interface: Records purchase, price, transport, and storage details on the blockchain, and linking these details to a QR attached to the produce.

Retailer Interface: Logs final sales, generates a QR code linked to blockchain records for consumer access.

Consumer Interface: Scans QR to view the full journey of the produce, including pricing, certifications, and quality checkpoints.

❖ INNOVATION & UNIQUENESS

CORE IDEA

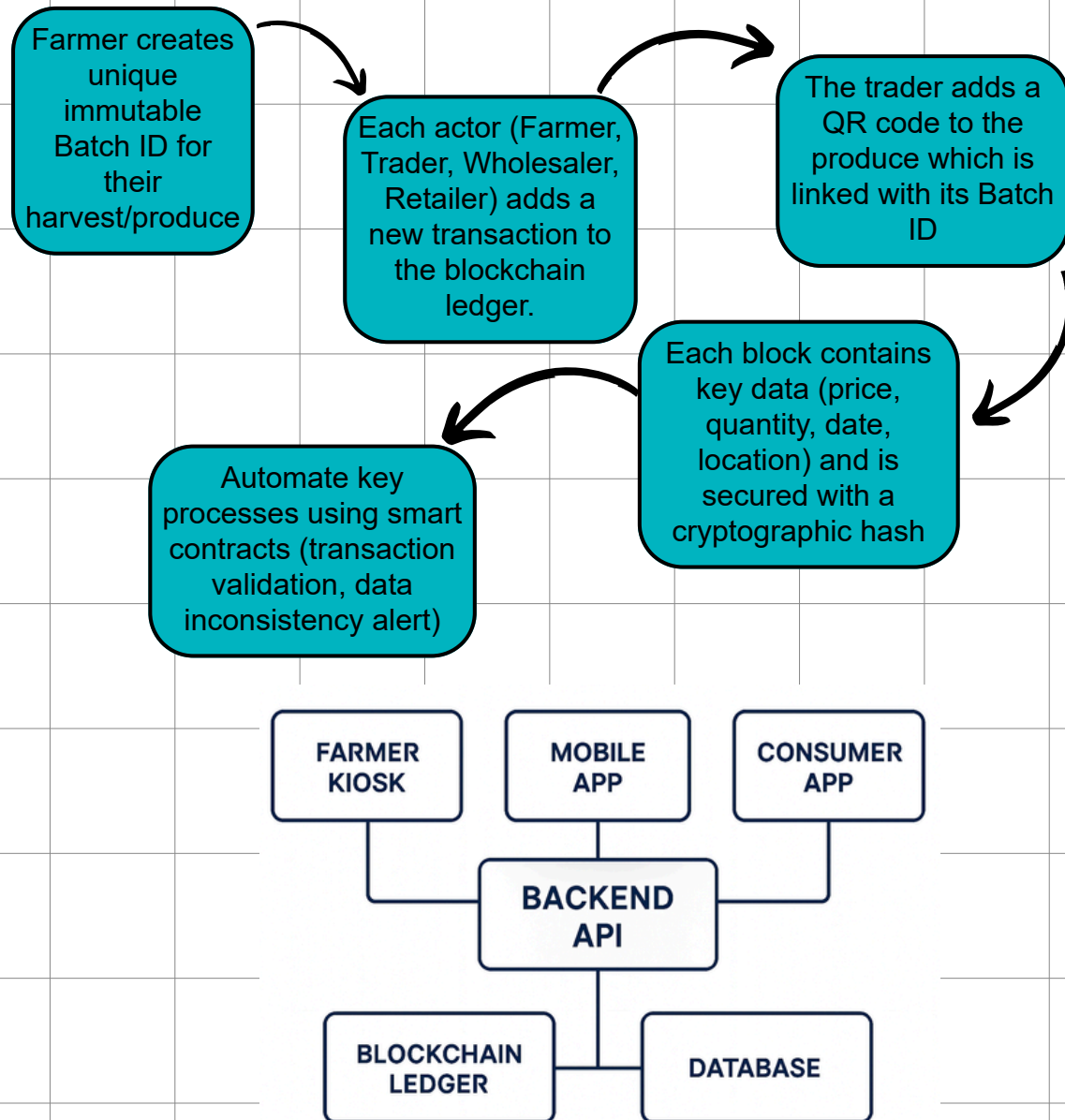
- **Blockchain Ledger:** Use a permissioned blockchain (like Hyperledger Fabric or Polygon) to create an unchangeable digital ledger of all transactions.
- **Smart Contracts:** Automate trust and enforce rules between all parties in the supply chain.
- **Low-Cost Interface:** Implement simple, accessible interfaces for each stakeholder—a mobile app for distributors, kiosks for farmers, and a static QR code for retailers.
- **QR code integration:** QR code integration makes tracing and verifying products easy.
- **UPI integration:** Online payments make it easy to track viable transactions.

Retailer-Level QR Code:
A single static QR code at the point of sale fits the Indian loose produce market, eliminating the need for expensive per-item barcoding.

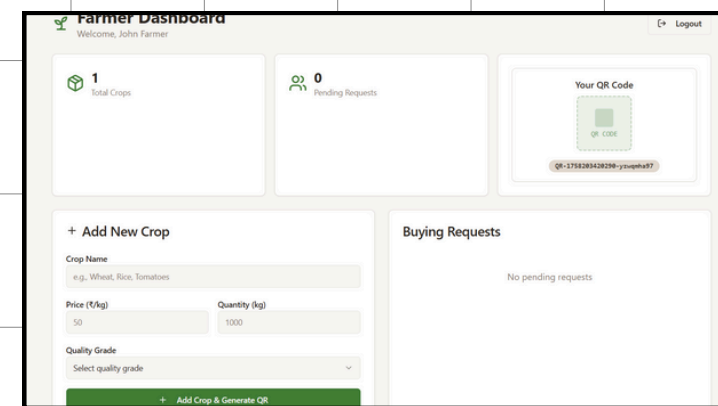
Decentralized system:
Everything is stored on blockchain which makes it impossible for middle men to manipulate prices.

Fair Price Awareness:
The system integrates MSP provided by the govt or current market at each levels.

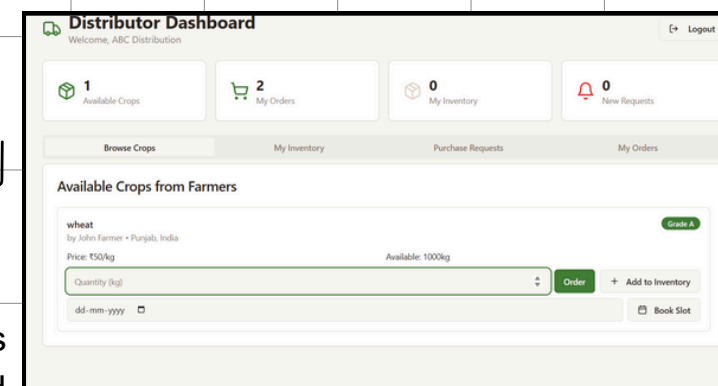


BLOCKCHAIN AND DATA FLOW**DATA INTEGRATION AND UI****FARMER INTERFACE**

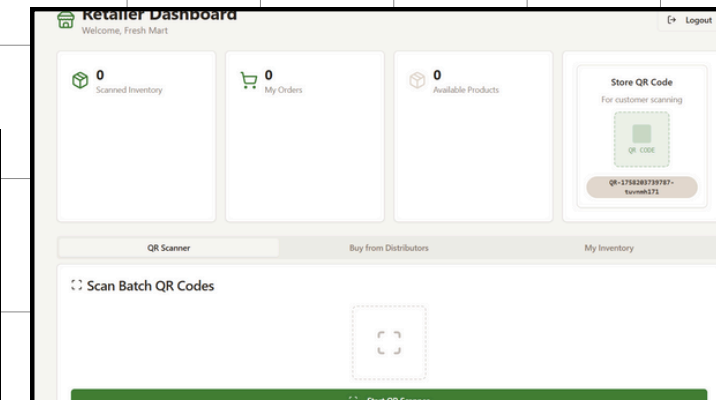
Simple kiosk based to overcome smartphone or tech-literacy barriers

**TRADER INTERFACE**

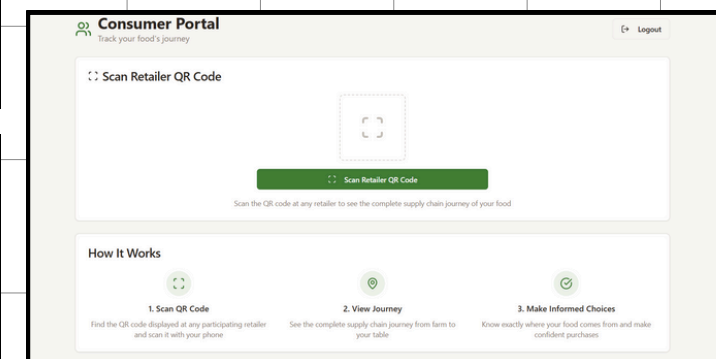
Transparency in transactions that would otherwise be hard to track for distributor

**RETAILER INTERFACE**

Retailers have a QR code displayed in their shop which gives all details of the retailer's inventory, with a blockchain record associated with every crop.

**CONSUMER INTERFACE**

When a consumer scans the retailer's QR, the app queries the blockchain to fetch and display the inventory of the retailer, which can be explored further for the desired product details.

**DEVELOPMENT METHODOLOGY**

We will follow an agile development methodology, focusing on a Minimum Viable Product (MVP) that covers the core traceability loop (farmer to consumer).

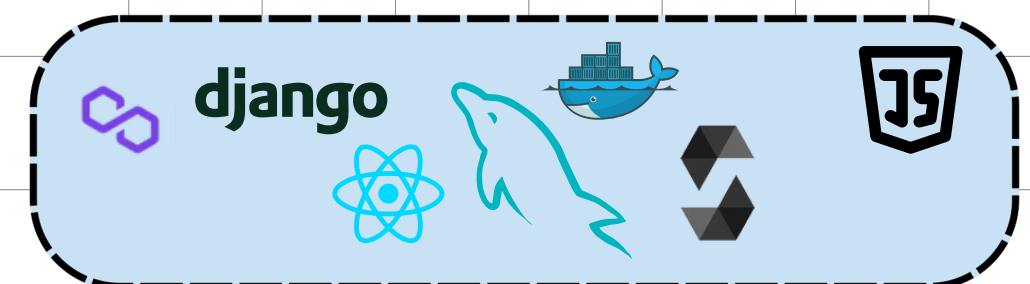
Requirement Analysis

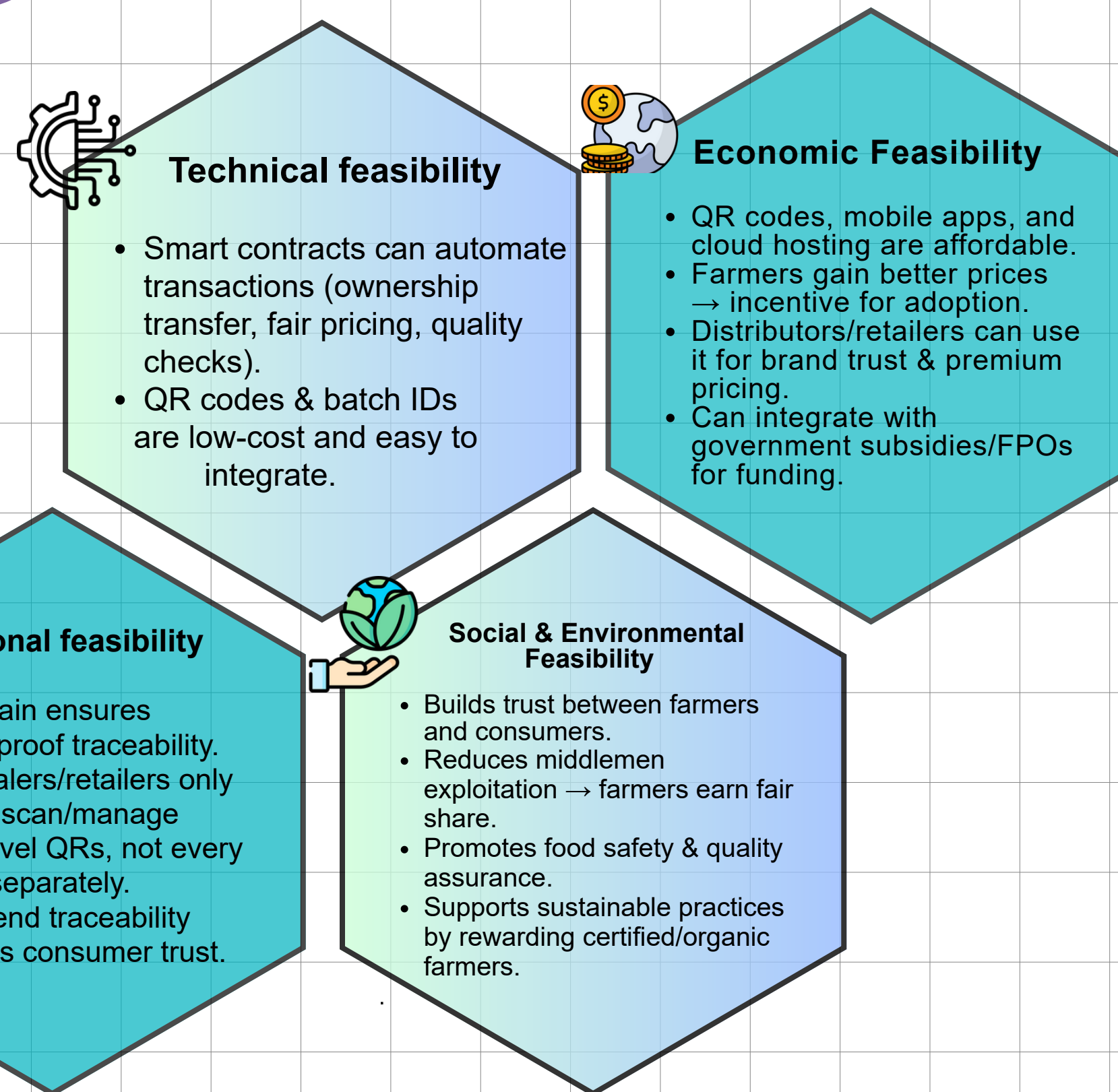
API and Database Integration

Testing and Deployment

Blockchain Setup and Smart Contract

UI/UX Design and Front-end Development

FRAMEWORKS & PROGRAMMING LANGUAGES



POTENTIAL CHALLENGES AND THEIR SOLUTIONS

1) Many farmers may not be comfortable using apps and smartphones.

- Provide a simple mobile app in regional languages with icons/voice inputs.
- Training and organising workshops for farmer's education.
- Kiosk setups for registering produce details on behalf of the farmer.

2) Middlemen may resist as blockchain reduces hidden margins.

- Position system as value-adding (better branding, trust, access to premium buyers).
- Setting up rating and review system for middlemen. Business gets attracted to high-rated distributors.
- Involve FPOs or govt. mandis first → private distributors join when they see benefits.

3) Produce often changes multiple hands (farmer → aggregator → wholesaler → retailer) anyone can degrade the quality..

- Tamper-proof seals & QR tags on farmer batches.
- At each stage, the buyer checks quality, puts "crop quality" as "Good", "Bad" or "Medium".
- Use batch-aggregation model → one new QR generated at collector stage (linked to farmer QRs)..
- Consumer app shows warning if quality data missing.
- Integrating shelf life of crops in our app.

IMPACT

Transparency: Every transaction (farmer → distributor → consumer) is recorded immutably.

Regulatory Compliance: Easier audits and government monitoring.

Trust Building: Consumers can verify product authenticity via QR code.

Farmer Empowerment: Direct visibility of fair pricing; reduced exploitation by middlemen.

Fraud Reduction: Tamper-proof records prevent fake invoices & quality reports.

Supply Chain Efficiency: Real-time tracking reduces delays, improves logistics.



Farmers get increased income and better pricing of their produce.



Distributors/Retailers get streamlined traceability.

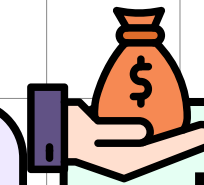


Consumers get assurance of food safety and trace product journey.



Social Benefits

- **Farmer Empowerment:** Fair pricing, reduced exploitation, direct recognition.
- **Consumer Trust:** Transparency in product journey and authenticity.
- **Community Development:** Stronger farmer–consumer connections.
- **Food Safety & Health:** Verified, unadulterated produce reaches families.



Economic Benefits

- **Fair Trade:** Farmers get better income, reducing rural poverty.
- **Efficiency:** Lower transaction costs, reduced paperwork, faster settlements.
- **Market Expansion:** Verified supply chains open up export opportunities.
- **Job Creation:** New roles in agri-tech, blockchain integration, logistics.
- **Reduced Fraud:** Prevents fake invoices, tampering, and middlemen exploitation.



Environmental Benefits

- **Reduced Food Waste:** Real-time tracking reduces spoilage during transit.
- **Sustainable Practices:** Verified certifications encourage organic and eco-friendly farming.
- **Efficient Logistics:** Optimized transport reduces carbon footprint.
- **Resource Optimization:** Better crop planning reduces overproduction and land misuse.

**PERSPECTIVE REQUIREMENTS
CONSIDERED**

- 1) **Farmer**
Requires transparency regarding prices at which their produce is being sold.
- 2) **Distributor + Retailer**
Require a tool for communicating inventory, price information, and produce freshness between one another
- 3) **Consumer**
Requires honest data regarding freshness of produce and place of origin



**LINKS AND
REFERENCES**

[Blockchain tech in supply chain management](#)

[Effect of Blockchain Technology on Supply Chain Sustainability Performances](#)

[Application of Blockchain to Supply Chain](#)

[Supply Chain Management and Indian Fresh Produce Supply Chain: Opportunities and Challenges](#)

[PROTOTYPE\(Link\).](#)

**FUTURE
OUTLOOK**

Technology Advancements

- AI-driven Insights: Integrate predictive analytics for crop yield forecasting, price trends, and demand forecasting to help farmers make data-driven decisions.

Business Growth

- Geographical Expansion: Scale from local to national coverage, eventually connecting with export markets.
- B2B Marketplace Integration: Enable partnerships with restaurants, food processors, and retailers for direct sourcing.

